

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		ratio C : H ⇒ $\frac{85.6}{12} : \frac{14.4}{1.01}$ ⇒ 7.13 : 14.26 (1) ratio = 1:2 ⇒ empirical formula is CH ₂ (1) <i>M_r</i> in the range 50-60 empirical formula <i>M_r</i> = 14 therefore molecular formula must be C ₄ H ₈ (1)					1	
		(ii)		0.2 mol of C ₄ H ₈ and 0.2 mol of bromine (1) so 1 double bond (1) must have some justification		1				
		(iii)		electrophilic addition	1			1		
		(iv)		radical substitution	1			1		
		(v)		compound X CH ₂ =CHCH ₂ CH ₃ or CH ₃ CH=CHCH ₃ (1) product of reaction of X with bromine in the absence of light CH ₂ BrCHBrCH ₂ CH ₃ or CH ₃ CHBrCHBrCH ₃ - must follow from part (i) (1) product of reaction of X with excess bromine in sunlight displayed formula of any substituted bromo-compound - must follow from part (ii) (1)						

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	(b)			Indicative content IR spectrum link C—H to 3000 and C—Cl to 650 Mass spectrum M_r is 126 / 128 / 130 two chlorine atoms present from peaks at 35 and 37 / peak height ratios / from M_r differences comment on any fragment ^{13}C NMR two different carbon environments comment on δ values ^1H NMR two different hydrogen environments same number of hydrogens in each environment $\Rightarrow$ 1,4-dichlorobutane rather than 2,3-dichlorobutane 5-6 marks At least one valid comment on all four spectra and a correct conclusion for the identity of Y (1,4-dichlorobutane) <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks At least one valid comment on three spectra and good attempt at the identity of Y. <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks At least one valid comment on two spectra . <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>		2	4	6		
				Question 8 total	2	6	8	16	1	0